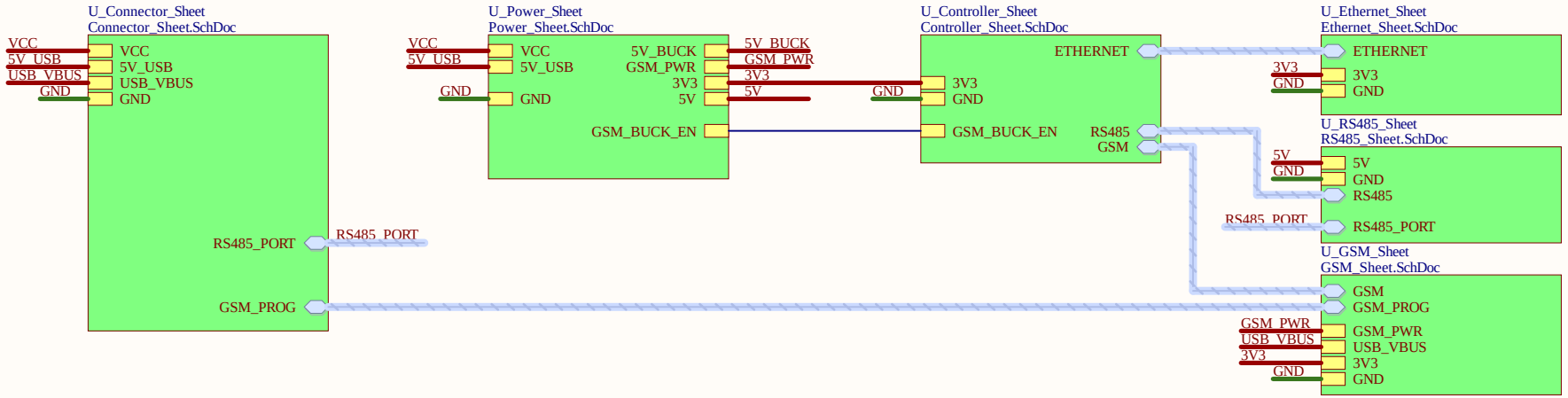
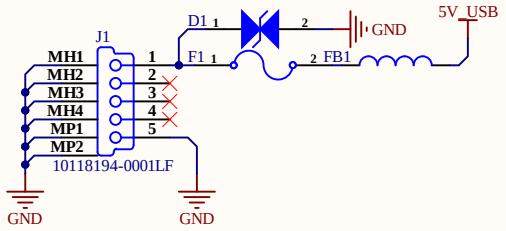
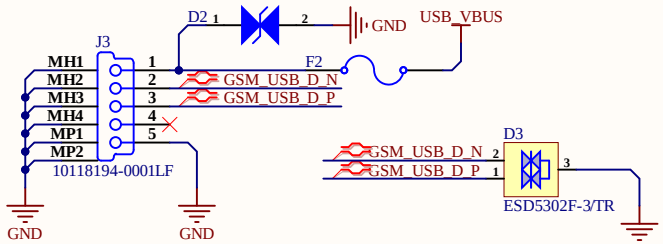


Title		
Size: A4	Number	Revision
Date: 4/24/2023		Sheet of
File: E:\Faizan\..\Top_Sheet.SchDoc		Drawn By:

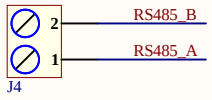
USB to Power Board



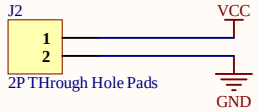
USB for GSM Updates



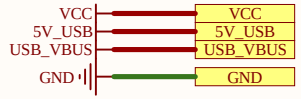
PORT for RS485

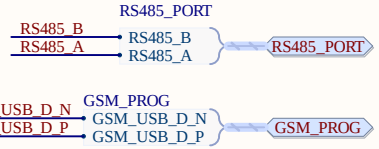


9-30V DC INPUT



Might need to change aftter confirmatioon

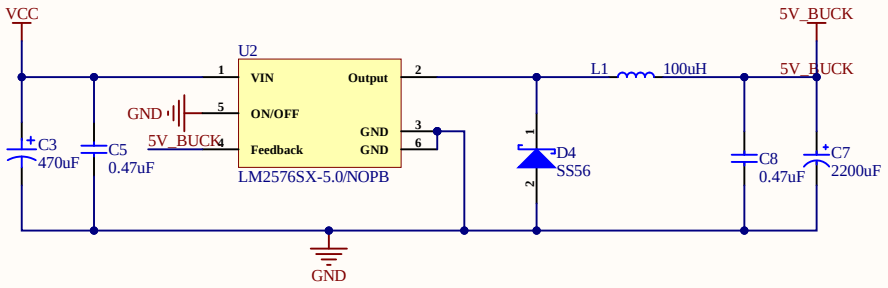




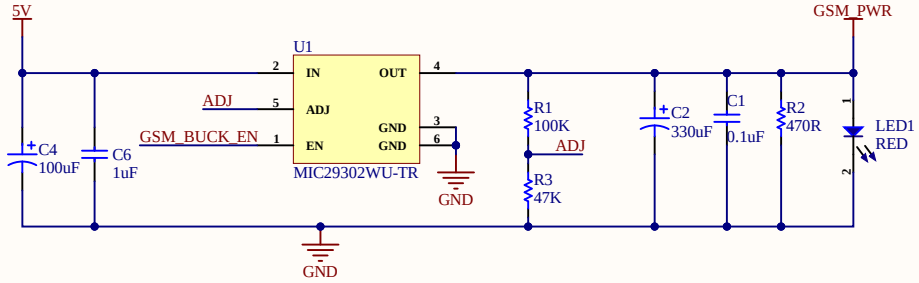
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Connector_Sheet.SchDoc	
File: ESP32IOTProduct.PrjPcb	
Date: 4/24/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 7

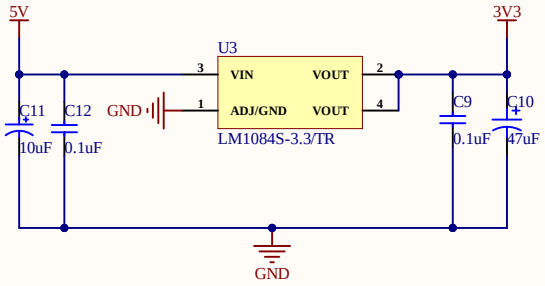
5V BUCK REGULATOR



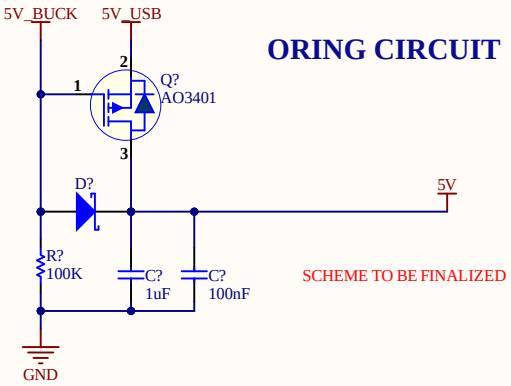
Buck for GSM Power

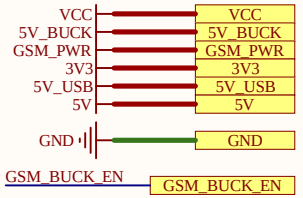


LDO 3.3V



ORING CIRCUIT



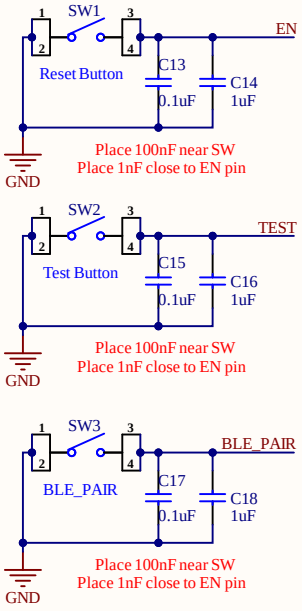


Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

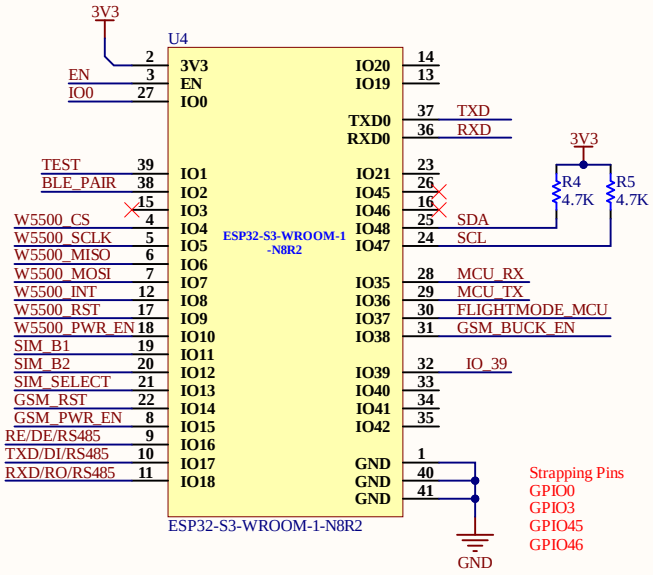
Title: Power_Sheet.SchDoc	
File: ESP32IOTProduct.PrjPcb	
Date: 4/24/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 7

CONTROLLER SECTION

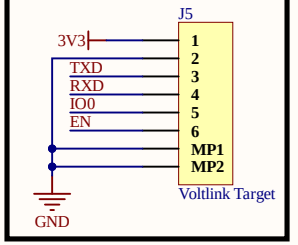
BUTTONS



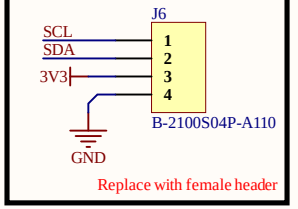
ESP32-S3

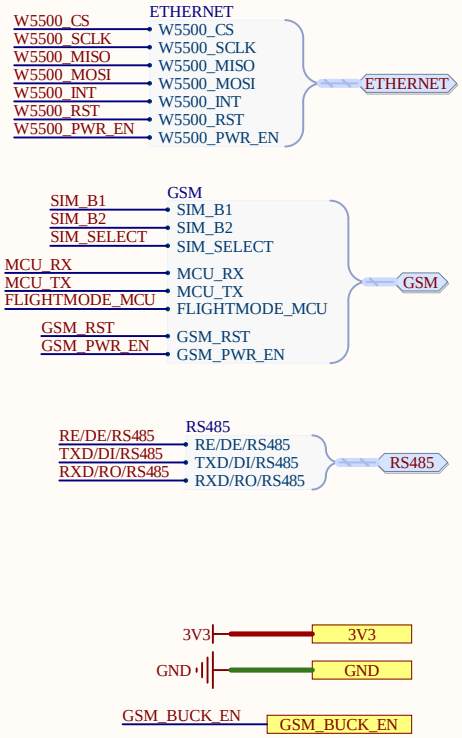


PROGRAMMING

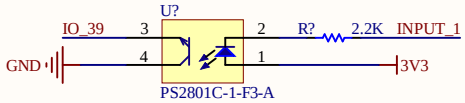


OLED HEADER





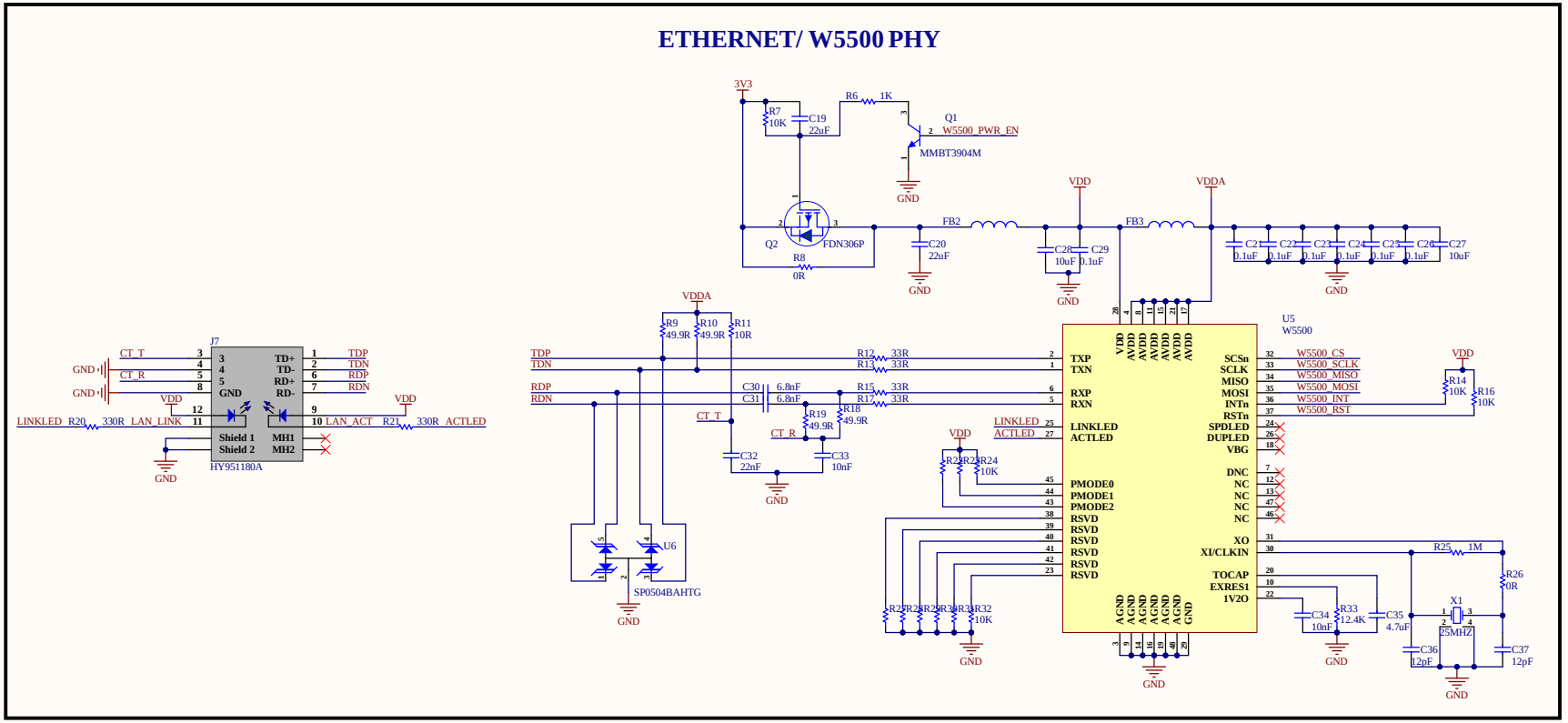
OPTO ISOLATION FOR GPIOs

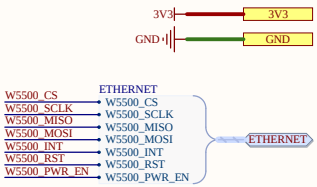


Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Controller_Sheet.SchDoc	
File: ESP32IOTProduct.PrjPcb	
Date: 4/24/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 7

ETHERNET/ W5500 PHY

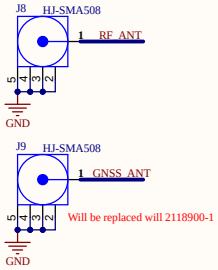




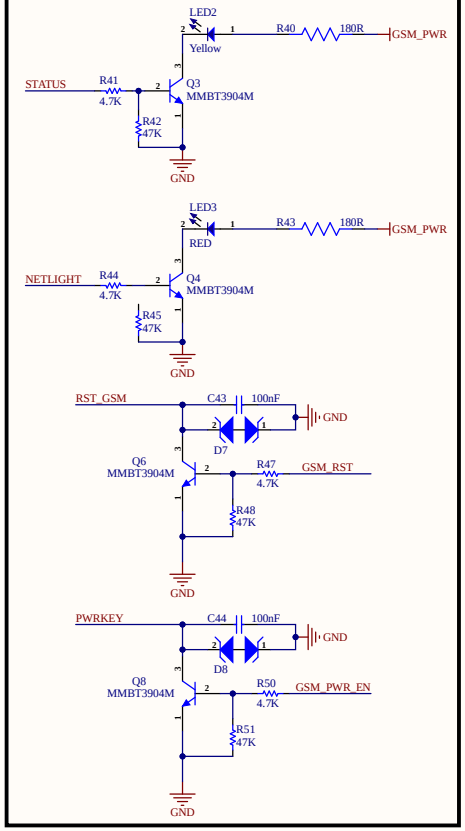
Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Ethernet_Sheet.SchDoc	
File: ESP32IOTProduct.PrjPcb	
Date: 4/24/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 7

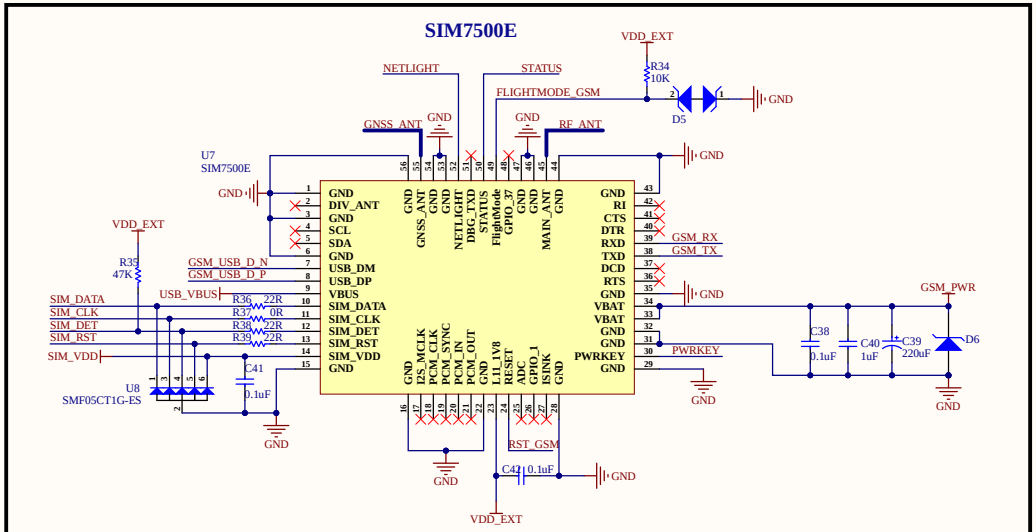
ANTENNA



GSM_COMMANDS



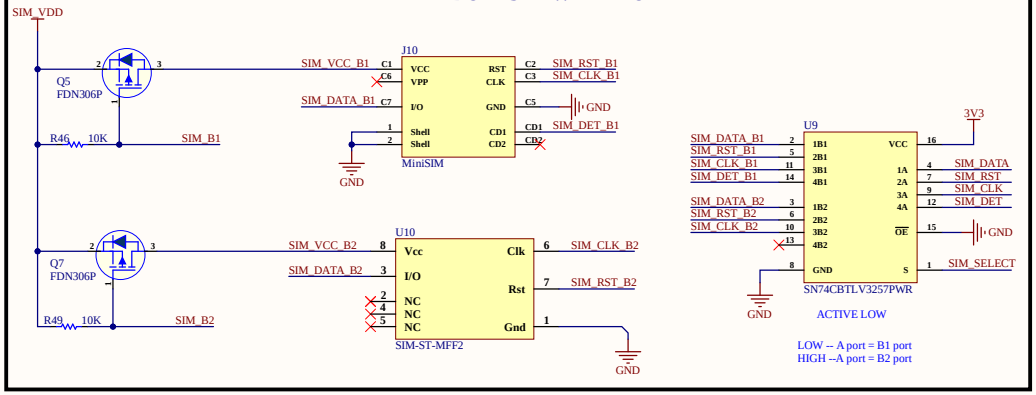
SIM7500E



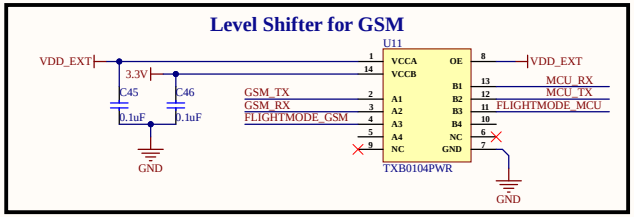
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

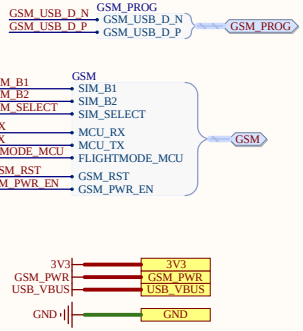
Datasheet Pg 31

DUALSIM WITH MULTIPLEXER



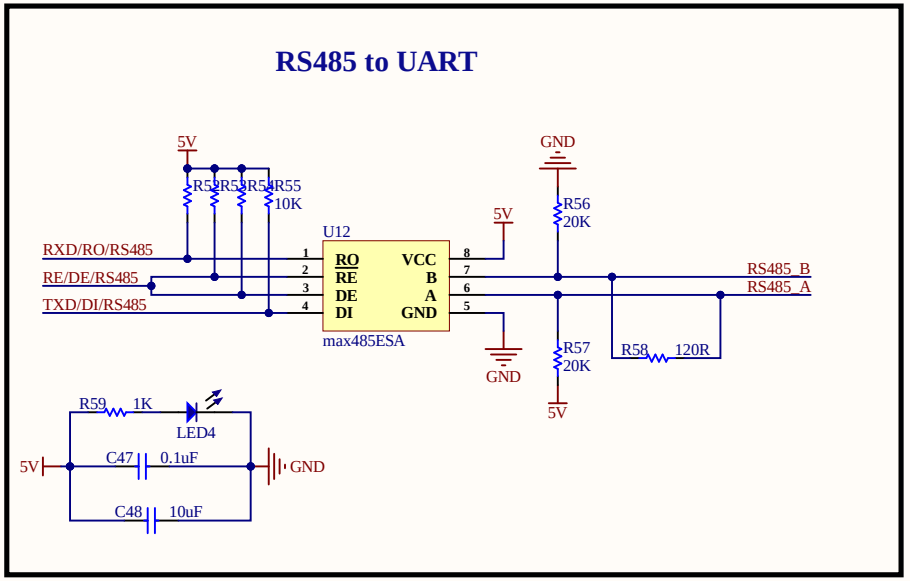
Level Shifter for GSM

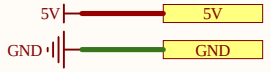


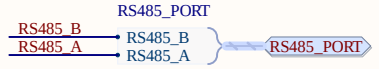


Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: GSM_Sheet.SchDoc	
File: ESP32IOTProduct.PrjPcb	
Date: 4/24/2023	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 7









Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: RS485_Sheet.SchDoc	
File: ESP32IOTProduct.PrjPcb	
Date: 4/24/2023	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 7